Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
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Practice Problem Set-2 Nov, 2026

Foundations of Probability

Q1. Three Events and Inclusion–Exclusion

A construction activity can be delayed by three causes: labor shortage (L), late material supply (M) and bad weather (W). The probabilities are $\Pr(L) = 0.20$, $\Pr(M) = 0.15$, $\Pr(W) = 0.25$, $\Pr(L \cap M) = 0.05$, $\Pr(L \cap W) = 0.06$, $\Pr(M \cap W) = 0.04$ and $\Pr(L \cap M \cap W) = 0.02$.

(a) Find the probability that the activity is delayed, and the probability that it is not delayed.
(b) Find the probability that the delay has exactly one cause.
(c) Find $\Pr(W \mid L)$. Are L and W independent?
(d) Find $\Pr(L \cup M \mid W)$.

Q2. Total Probability and Bayes' Rule

The condition of a bridge is classified as Good (G), Fair (F) or Poor (P), with prior probabilities 0.70, 0.25 and 0.05. A monitoring system gives an alarm with probability 0.02, 0.10 and 0.80 when the condition is G, F and P, respectively.

(a) Find the probability of an alarm.
(b) Given an alarm, find the posterior probabilities of G, F and P.
(c) Given no alarm, find the posterior probabilities of G, F and P.
(d) Two readings are taken. Given the condition, the readings are independent. Both readings give an alarm. Find the probability that the condition is P.
(e) Explain why the posterior probability of P in (b) is much smaller than 0.80.

Random Variables and Probability Models

Q3. Repeated Independent Trials

Concrete cubes are tested one at a time. Each cube fails the test independently with probability 0.1.

(a) Find the probability that exactly 2 of 10 cubes fail.
(b) Find the probability that at most 1 of 10 cubes fails.
(c) Find the mean and variance of the number of failures among 10 cubes.
(d) Find the probability that the first failure occurs at the 5th cube.
(e) Find the probability that the second failure occurs at the 8th cube.
(f) Find the expected number of cubes tested up to and including the first failure.

Q4. Poisson Occurrences

Earthquakes of magnitude 5 or more occur in a region as a Poisson process with a mean rate of 0.15 per year.

(a) Find the probability of no such earthquake in 10 years.
(b) Find the probability of at least two such earthquakes in 10 years.
(c) Find the mean and variance of the number of such earthquakes in 20 years.
(d) Find the probability that the time to the next such earthquake exceeds 5 years.
(e) Find the period t (years) for which the probability of at least one such earthquake is 0.5.

Q5. Function of a Random Variable

The wind speed V (m/s) is uniformly distributed between 20 and 30 m/s. The wind pressure is $P = 0.6\,V^2$ (N/m^2).

(a) Find the range of P.
(b) Find the CDF of P using the CDF method.
(c) Find the PDF of P.
(d) Find $E[P]$, and compare it with $0.6\,(E[V])^2$.
(e) Find $\Pr(P > 450$ N/m$^2)$.

Normal and Lognormal Distributions

Q6. Normal Probabilities and Percentiles

The 28-day compressive strength X of concrete cubes from a plant is normally distributed with mean 32 MPa and standard deviation 4 MPa.

(a) Find $\Pr(X < 25$ MPa).
(b) Find $\Pr(28 < X < 36$ MPa).
(c) Find the 5th percentile of X.
(d) Find the probability that the average of 3 independent cubes is below 28 MPa.
(e) The standard deviation stays at 4 MPa. Find the mean strength needed so that the 5th percentile is at least 30 MPa.

Q7. Products of Lognormal Variables

The duration D (days) of a construction activity is lognormal with mean 20 days and coefficient of variation 0.25. The cost rate U (thousand rupees per day) is lognormal with median 50 and $\zeta = 0.20$, and is independent of D. The total cost is $K = U\,D$.

(a) Find λ_D, ζ_D and the median of D.
(b) Find $\Pr(D > 25$ days).
(c) Show that K is lognormal, and find λ_K, ζ_K and the median of K.
(d) Find $\Pr(K > 1300)$ (thousand rupees).
(e) Find the mean of K, and explain why it differs from the median.

Multivariate Probability and Uncertainty Propagation

Q8. Joint PMF of Two Discrete Variables

X and Y are the numbers of lane closures on two adjacent bridges on a given day. Their joint PMF is:

$p_{XY}(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	0.20	0.10	0.02
$x=1$	0.10	0.20	0.08
$x=2$	0.02	0.08	0.20

(a) Find the marginal PMFs of X and Y.
(b) Are X and Y independent?
(c) Find $E[X]$, $\mathrm{Var}(X)$, $\mathrm{Cov}(X,Y)$ and ρ_{XY}.
(d) Find the conditional PMF of Y given $X = 1$, and $E[Y \mid X = 1]$.
(e) Find $\Pr(X + Y \geq 3)$.

Q9. First- and Second-Order Approximations
The midspan deflection of a beam is $\delta = 150/E$ (mm), where the elastic modulus E (GPa) has mean 30 GPa and standard deviation 3 GPa.

(a) Find the first-order mean and standard deviation of δ.
(b) Find the second-order mean of δ.
(c) The stress in a bar is $\sigma = P/A$. The load P has mean 200 kN and standard deviation 20 kN; the area A has mean 0.01 m^2 and standard deviation 0.0005 m^2; P and A are independent. Find the first-order mean and standard deviation of σ (MPa).
(d) Show that the coefficient of variation of σ in (c) is approximately $\sqrt{\delta_P^2 + \delta_A^2}$, and check it with the numbers.

Central Limit Theorem and Extremes

Q10. Central Limit Theorem
The mass of a concrete block has mean 25 kg and standard deviation 1.2 kg. Its distribution is not known. Masses of different blocks are independent.

(a) A pallet holds 100 blocks. Find the mean and standard deviation of the total mass.
(b) Find, approximately, the probability that the total mass exceeds 2520 kg.
(c) For a random sample of 36 blocks, find, approximately, the probability that the sample mean is below 24.6 kg.
(d) Find the smallest sample size n for which $\Pr(|\overline{X} - 25| < 0.2 \text{ kg}) \geq 0.95$.
(e) Explain why the normal distribution can be used in (b)–(d).

Q11. Gumbel Distribution Fitted by the Method of Moments
Annual maximum wind speeds V at a station have a sample mean of 34 m/s and a sample standard deviation of 6 m/s. Assume V follows a Gumbel (Type I, largest) distribution with CDF $F_V(v) = \exp[-\exp(-\alpha(v-u))]$, mean $u + 0.5772/\alpha$ and standard deviation $\pi/(\alpha\sqrt{6})$.

(a) Estimate u and α by MoM.
(b) Find the probability that V exceeds 45 m/s in a given year.
(c) Find the 50-year return value of V.
(d) Show that the maximum of V over n independent years is also Gumbel, with the same α and $u_n = u + (\ln n)/\alpha$.
(e) Find the probability that the maximum wind speed over 20 years is below 45 m/s.

Parameter Estimation and Confidence Intervals

Q12. MoM and MLE for a Uniform Distribution
Waiting times X (minutes) at a traffic signal are uniformly distributed on $(0, \theta)$. A sample of 8 waiting times is: 3.1, 7.4, 1.2, 5.6, 8.9, 2.5, 4.0, 6.3.

(a) Find the MoM estimator of θ and its value.
(b) Write the likelihood function $L(\theta)$, and show that the MLE of θ is the sample maximum.
(c) Find the MLE of θ.
(d) Given $E[\max X_i] = n\theta/(n+1)$, state whether the MLE is biased. Give an unbiased estimator based on the maximum, and its value.
(e) For the sample 1, 1, 1, 9, compute the MoM estimate of θ and comment.

Q13. Confidence Intervals from Data

The unit weights (kN/m^3) of 8 soil samples are: 18.2, 17.6, 18.9, 18.4, 17.9, 18.6, 18.1, 18.3. Assume a normal population.

(a) Find the sample mean and sample standard deviation.
(b) Construct a 95% two-sided confidence interval for the mean μ.
(c) Construct a 95% lower confidence bound for μ.
(d) Construct a 95% two-sided confidence interval for the standard deviation σ.
(e) Assume $\sigma = 0.4$ kN/m^3 is known. Find the sample size needed for a 95% two-sided interval of half-width 0.2 kN/m^3.

Hypothesis Testing

Q14. One-Sample z-Test and Power

The specification for a compacted subgrade requires a mean CBR of at least 8%. The standard deviation of CBR is known to be 1.5%. Nine tests give a sample mean of 7.2%.

(a) Test $H_0 : \mu = 8\%$ against $H_1 : \mu < 8\%$ at $\alpha = 0.05$, and find the p-value.
(b) Find the critical value of the sample mean for this test.
(c) Find the probability of a Type II error if the true mean is 7%, and the power of the test.
(d) Find the sample size needed for the Type II error probability at a true mean of 7% to be at most 0.10 (with $\alpha = 0.05$).
(e) State the conclusion of (a) in words.

Q15. Two-Sample z-Test and Confidence Interval

The dry density (g/cm^3) of a soil compacted by two methods is compared. Method-1: $n_1 = 12$, sample mean 1.82, known standard deviation 0.06. Method-2: $n_2 = 16$, sample mean 1.77, known standard deviation 0.08.

(a) Test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$ at $\alpha = 0.05$.
(b) Find the p-value of the test in (a).
(c) Test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 > \mu_2$ at $\alpha = 0.05$.
(d) Construct a 95% two-sided confidence interval for $\mu_1 - \mu_2$, and relate it to (a).

Q16. Paired Measurements

The deflections (mm) of six beams were measured with two instruments, Gauge-A and Gauge-B:

Beam	1	2	3	4	5	6
Gauge-A	12.4	10.8	15.2	9.6	13.1	11.5
Gauge-B	12.0	10.1	14.9	9.4	12.5	11.4

(a) Find the differences (A $-$ B), their mean and their standard deviation.
(b) Test, at $\alpha = 0.05$ (two-sided), whether the two gauges give the same mean reading.
(c) Treat the two columns as independent samples with equal variances, and repeat the test.
(d) Explain why the conclusions differ, and which test is appropriate.

Q17. Tests on the Mean and the Variance

Twenty cubes of a new concrete mix have a sample mean strength of 41.8 MPa and a sample standard deviation of 3.6 MPa. The specified mean is 40 MPa, and the historical standard deviation is 5 MPa. Assume a normal population.

(a) Test, at $\alpha = 0.05$, whether the mean strength exceeds 40 MPa.
(b) Give the range within which the p-value of (a) lies.
(c) Test, at $\alpha = 0.05$, whether the standard deviation is less than 5 MPa.
(d) Repeat (c) with a two-sided alternative, and compare the conclusions.
(e) Construct a 95% two-sided confidence interval for the standard deviation.

Regression

Q18. Least-Squares Fit from Data

The traffic volume x (thousand vehicles per day) and the noise level y (dB) at six locations are:

x	2	4	6	8	10	12
y	61	64	66	70	71	75

(a) Fit the least-squares line $\hat{y} = \hat{\alpha} + \hat{\beta}x$.
(b) Find the residuals, and check that they sum to zero.
(c) Find the conditional standard deviation $s_{Y|x}$.
(d) Construct 95% confidence intervals for the mean noise level at $x = 7$ and at $x = 12$. Compare their widths.
(e) Assume Y at a given x is normal, with mean on the fitted line and standard deviation $s_{Y|x}$. Find $\Pr(Y > 76 \text{ dB} \mid x = 12)$.

Q19. Least Squares in Matrix Form

Four observations are $(x, y) = (0, 1.1)$, $(1, 2.9)$, $(2, 5.2)$, $(3, 6.8)$. Fit $y = \alpha + \beta x$.

(a) Write the matrix A and the vector $\mathbf{y}$ for the system $A\,[\alpha\ \beta]^T \approx \mathbf{y}$.
(b) Compute A^TA and $A^T\mathbf{y}$.
(c) Solve $[\hat{\alpha}\ \hat{\beta}]^T = (A^TA)^{-1}A^T\mathbf{y}$.
(d) Check the result with the scalar formulas for $\hat{\beta}$ and $\hat{\alpha}$.
(e) Find the fitted values, the residuals and the residual sum of squares.

Reliability Analysis

Q20. Failure Probability by Conditioning on a Discrete Demand

The capacity R of a member is normally distributed with mean 150 kN and standard deviation 15 kN. The demand S is independent of R and takes the values 100, 120 and 140 kN with probabilities 0.5, 0.3 and 0.2.

(a) Write the limit-state function and the failure event.
(b) Find the failure probability by conditioning on S.
(c) Given that the member has failed, find the probability that $S = 140$ kN.
(d) Find the central safety factor based on the mean demand, and comment on what it does not show.
(e) The cost of failure is 50 lakh rupees. Find the expected loss (risk).

Q21. Failure Probability by Conditioning on a Continuous Capacity

The capacity R is uniformly distributed between 80 and 120 kN. The demand S is exponentially distributed with mean 40 kN, and is independent of R.

(a) Write the limit-state function and sketch the failure domain in the (r, s) plane.
(b) Write the failure probability as an integral by conditioning on R.
(c) Evaluate the failure probability.
(d) Find the reliability.
(e) Find the failure probability if R is fixed at 100 kN, and compare with (c).

Q22. Lognormal Capacity and Demand

The capacity R is lognormal with median 300 kN and $\zeta_R = 0.15$. The demand S is lognormal with median 150 kN and $\zeta_S = 0.30$. R and S are independent.

(a) Find the distribution of $\ln R - \ln S$, the reliability index β and the failure probability.
(b) Find the ratio of median capacity to median demand needed for $\beta = 3.0$ (same ζ_R and ζ_S).
(c) Find the means and standard deviations of R and S. Using these, compute β as if R and S were normal.
(d) Compare the failure probabilities from (a) and (c), and comment.

Q23. Monte Carlo Simulation

The failure probability for the limit-state function $g(R, S) = R - S$ is to be estimated by Monte Carlo simulation.

(a) List the steps of the simulation.
(b) Define the indicator I_i for the i-th sample, write the estimator $\hat{p}_f$ from N samples, and show that $E[\hat{p}_f] = p_f$.
(c) Derive $\mathrm{Var}(\hat{p}_f)$ and the coefficient of variation of $\hat{p}_f$.
(d) For $p_f \approx 10^{-3}$, find the number of samples needed for a coefficient of variation of 0.10.
(e) A run with $N = 20{,}000$ gives 30 samples with $g \leq 0$. Find $\hat{p}_f$ and its estimated coefficient of variation.
(f) The integral $\int_0^1 e^{-x^2}\,dx$ is to be estimated by simulation. Ten uniform random numbers on $(0, 1)$ are 0.12, 0.85, 0.47, 0.33, 0.91, 0.64, 0.05, 0.72, 0.29, 0.58. Find the estimate.

■

Standard Normal Probabilities

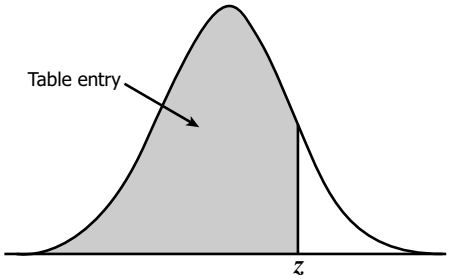
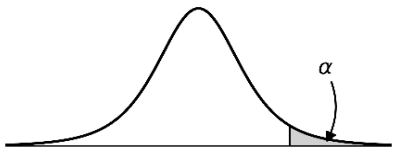
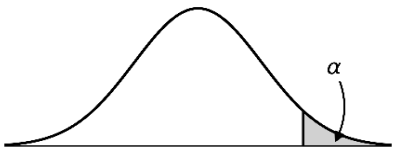


Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

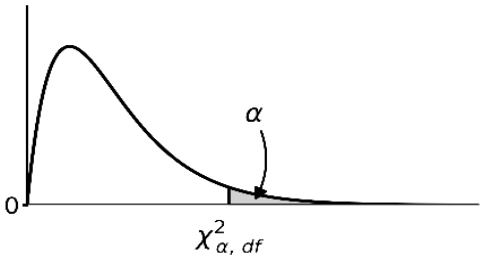
t Table





cum. Prob	$t_{0.50}$	$t_{0.75}$	$t_{0.80}$	$t_{0.85}$	$t_{0.90}$	$t_{0.95}$	$t_{0.975}$	$t_{0.99}$	$t_{0.995}$	$t_{0.999}$	$t_{0.9995}$
one-tail (α)	**0.5**	**0.25**	**0.2**	**0.15**	**0.1**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	**0.000**	**0.674**	**0.842**	**1.036**	**1.282**	**1.645**	**1.960**	**2.326**	**2.576**	**3.090**	**3.291**
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level (two-tailed)										

Note: For df larger than 100, use the 1000 value; for larger than 1000, use the Z value



χ^2 Table

df	$\chi^2_{0.995}$	$\chi^2_{0.990}$	$\chi^2_{0.975}$	$\chi^2_{0.950}$	$\chi^2_{0.900}$	$\chi^2_{0.100}$	$\chi^2_{0.050}$	$\chi^2_{0.025}$	$\chi^2_{0.010}$	$\chi^2_{0.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169